SHRED-MPC loop

MODEL PREDICTIVE CONTROLLER

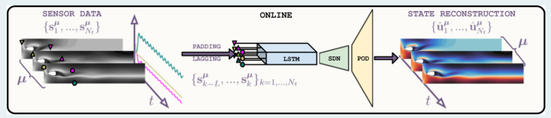
INPUT SENSORS

- Sparse pressure
  $\overline{p_s}(t-L+1 : t)$
- Rotational velocities
  $\overline{\omega}(t-L+1 : t)$

L = lag length (timesteps)

SHRED SURROGATE

Control-conditioned predictor



State space reconstruction and loss evaluation

OPTIMIZER

Minimize total cost

Candidate rotational velocity

PHYSICAL PLANT

OpenFOAM CFD

Advance the flow and return new sparse pressure data

At each MPC update: predict many candidate futures with SHRED, apply one control segment in CFD, then replan.

SHRED-ROM online panel (cropped): Matteo Tomasetto – Stefano Riva (2024), MIT. Upstream architecture illustration.